# Risk management for options trading

## Algorithmic Trading & Options Risk Management

9 April 2026

Speaker

**Dr. Gaurav Raizada**

## About The Speaker

### Dr. Gaurav Raizada

Co-Founder, iRage & QuantInsti | Visiting Professor, IIM Ahmedabad

Dr Gaurav Raizada is a Co-founder of iRage & QuantInsti, he is focused on developing trading strategies, optimising systems, and reducing latency. His earlier role as a derivatives trader at Axis Bank gave him in-depth knowledge of Forex and interest rate markets.

Dr Raizada holds a Ph.D. in Financial Econometrics from IIT Bombay, along with qualifications from IIM Lucknow and IIT Kanpur. His research includes studies on market efficiency and statistical anomalies in financial markets.

# The institutional risk mindset

## How is risk perceived

In professional trading, risk is often seen as a source of edge.

## The limits of forecasting

You cannot predict price reliably, but you can control exposure.

## The professional mandate

Manage the downside; the upside takes care of itself.

## Practitioner insight

We often see strategies that look perfect in a backtest fail during a 'Black Swan' event. From your experience at iRage, what is the biggest gap between theoretical risk models and and how microstructure reality behaves during a tail-risk crisis?

# What Actually Constitutes a Trading Edge?

❌ **NOT an Edge:**

- Time decay (theta) into expiration
- Well-known, priced-in, universally accessible

✅ **Real Edges:**

Technology Edge: Superior infrastructure, faster execution

Size Edge: Ability to post large bids/offers on illiquid options retail can't access

Information Edge: Proprietary data or insights

Pricing Edge: Better models, more accurate valuation

**Four Types of Alpha:**

1. Model Alpha – Superior pricing/risk models
2. Speed Alpha – Faster execution & reaction
3. Information Alpha – Exclusive data/insights
4. Size Alpha – Capacity to move/access markets others can't or Vice Versa

**Key Insight: If everyone knows it and can do it, it's not an edge, it's just market mechanics.**

# The Starting Strategy: Selling Overpriced Puts

Setup- Weekly 7 DTE puts @ ~15 Delta

Initial Hypothesis - Small position sizing + passive management = viable long-term edge

Risk Guardrail - Strict size limits to survive ~5-7% market drawdowns without blowing up

Theoretical Edge - Captures Variance Risk Premium (VRP) → terminal positive expectancy

Reality Check - Positive long-term expectancy ≠ smooth equity curve; path risk & drawdowns are inevitable

**Bottom Line - The math works over time, but real-world survival hinges on position sizing & drawdown tolerance.**

# Baking Reality Into Your Model: Execution, Slippage & Expectancy

**Core Principle**: If it can go wrong in live trading, it must be baked into your backtest assumptions.

📊 **Strategy Context (Example):**

- Sell NIFTY puts | 90 DTE | ~15 Delta | Nearly Reasonable liquidity (*Quote not Hit*)
- High DTE → Lower gamma → More stable, manageable risk profile

⚠️ **Execution Reality Check:**

- Slippage Example: Sell for Rs. 10.00 → Buy back at Rs 30.00–32.00
(10–20% relative credit impact)
- Impact: ↑ Required win rate to break-even OR ↓ Profit/Contribution Ratio (PCR) at same win rate
- Gap Risk: Events like Apr 8 and Apr 9 can cause 3–4x expected losses → must be modeled

# Baking Reality Into Your Model: Execution, Slippage & Expectancy

🔧 **Modeling Best Practices:**

1. Reject naive backtests – Execution assumptions are often overly optimistic
2. Use live fill data – Real-world slippage & execution quality inform realistic P&L
3. Stress-test for gaps – Incorporate tail-event multipliers into loss assumptions
4. Update continuously – More live samples = sharper, more robust expectancy models

🎯 **Strategic Framework (Across All Strategies):**

- Net premium selling + disciplined risk management
- Probability-driven position sizing & exit rules
- Expectancy = ƒ(win rate, payout ratio, slippage, gap risk, frequency)

**Bottom Line: A model is only as good as its assumptions. Bake in friction, gaps, and slippage before you trust the output.**

# Expected Value: Why Win Rate Alone Is a Trap

🎯 **The Win Rate Illusion:**

- *"I can build a 100% win rate strategy: Buy now, sell when profitable."*
- ✅ Technically true | ❌ Practically useless if you blow up on the 2% of trades that go against you
- **Key Question:** Can you survive the drawdown, not just win the trade?

Expected Value = (Win Rate × Avg Win) – (Loss Rate × Avg Loss) – Slippage – Gap Risk

*All inputs matter. Optimize the equation, not just one variable.*

Build for *robust expectancy*, not perfect win rate. Model slippage, gaps, hedge imperfections, and path risk—then size positions so you survive the outliers.

# Structuring options strategies systematically

## From bets to exposures

An options strategy is a deliberate exposure to specific risk factors (delta, gamma, theta, vega).

## Quantifying the objective

Systematic structuring requires isolating the desired Greek while hedging the unintended ones.

## The probability distribution of outcomes

Professional payoff diagrams are designed around a "range of outcomes" rather than a single target price.

## Practitioner insight

How does the underlying limit order book liquidity and the behavior of market makers influence the way we should view Greek risk?

# Execution and the hidden costs of trading

## The 'invisible tax' on alpha

Backtests often show profits that evaporate in live markets due to slippage, bid-ask spreads, and transaction costs.

## Market impact and liquidity

Large orders or illiquid option contracts can move the market against you.

## The latency of decision making

In options, the cost of being "slow" to adjust a hedge can exceed the cost of the trade itself.

## Practitioner insight

How does execution latency turn a theoretically 'delta-neutral' position into a directional gamble, and how should a trader quantify the 'market impact' of their own hedging activity?

# Zero DTE: Trading Gamma for Gap Risk

🎯 **The Core Trade-Off • You Gain:**

- No overnight exposure → Zero gap risk
- Intraday clarity → Clean daily reset
- Defined risk via spreads

• **You Accept:**

- Higher gamma → P&L moves faster & more violently
- Must sell closer to ATM → More directional sensitivity
- Vega hedges ineffective → No time to "wait out" volatility

# Zero DTE: Trading Gamma for Gap Risk

🛡️ **Risk Management Framework (Layered Defense)**

**1. Structure First**

- All positions are *spread off* → Fixed, known max loss per unit
- No naked short options → Eliminates undefined risk tail

**2. Stop Losses (Primary Tactical Defense)**

- Pre-defined exit levels (e.g., 2–3x credit received)
- Designed to cut losses early in adverse intraday moves

**3. Position Sizing (Ultimate Strategic Backstop)**

- Size for survivability, not peak optimization
- Assume stops may fail: slippage, EMP risk, tech outage
- Target outcome: "Lose a few weeks of profit" ≠ "Blow up the account"

# Zero DTE: Trading Gamma for Gap Risk

⚠️ **EMP Risk (Extreme Market Pathology)**

- Exchange outage / latency spike / inability to react
- Short-dated options offer zero time to recover or adjust
- **Only position sizing protects you when systems or markets fail**

🔑 **Key Insight** Zero DTE isn't "safer", it's *differently risky*. You swap gap risk for gamma risk and hedge fragility. The edge comes from disciplined structure + conservative sizing, not from the expiration itself.

**Bottom Line:** *When you can't hedge it and can't always exit it, you must size for it.*

# Avoiding the account blow-up (common pitfalls)

## The limitations of stop-loss

In volatile markets, a standard stop-loss may not execute at your price.

## Correlation breakdown

In a crisis, all assets tend to move together. A "diversified" portfolio of options can suddenly become a single, massive directional bet.

## ‘Unintended’ Greeks

Assignment risk, dividend risk, and "pin risk" at expiry can turn a winning trade into a capital-clearing event overnight.

## Practitioner insight

In low-liquidity environments, is a standard stop-loss giving us a false sense of security? How does 'slippage' change risk math for an algorithmic trader?

# Risk management as an edge

## Survival is a competitive advantage

In a zero-sum game, the "edge" belongs to the trader who can withstand the most volatility without forced liquidation.

## The quirks in compounding

A 50% loss requires a 100% gain to break even. Avoiding "left-tail" events is mathematically more important than chasing "right-tail" gains.

## The path to professionalism

Moving from "trading" to "quantitative asset management" requires a shift from discretionary intuition to a data-driven, risk-first infrastructure.

## Practitioner insight

One piece of advice for the individual trader moving to systematic options.

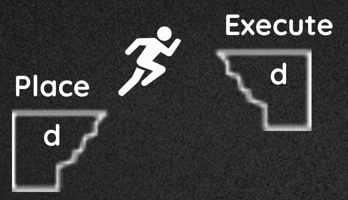
arrow.trade

# *Trade at the Speed of Light*

Retail Trading Platform with HFT Infrastructure

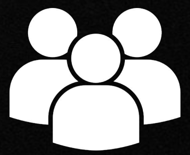
# The Retail Trading Landscape is Broken

**Large slippage**




**Technology debt**

**Built for Access/Everyone**



# With Arrow,

# Speed is your edge.

**Improved PnL**

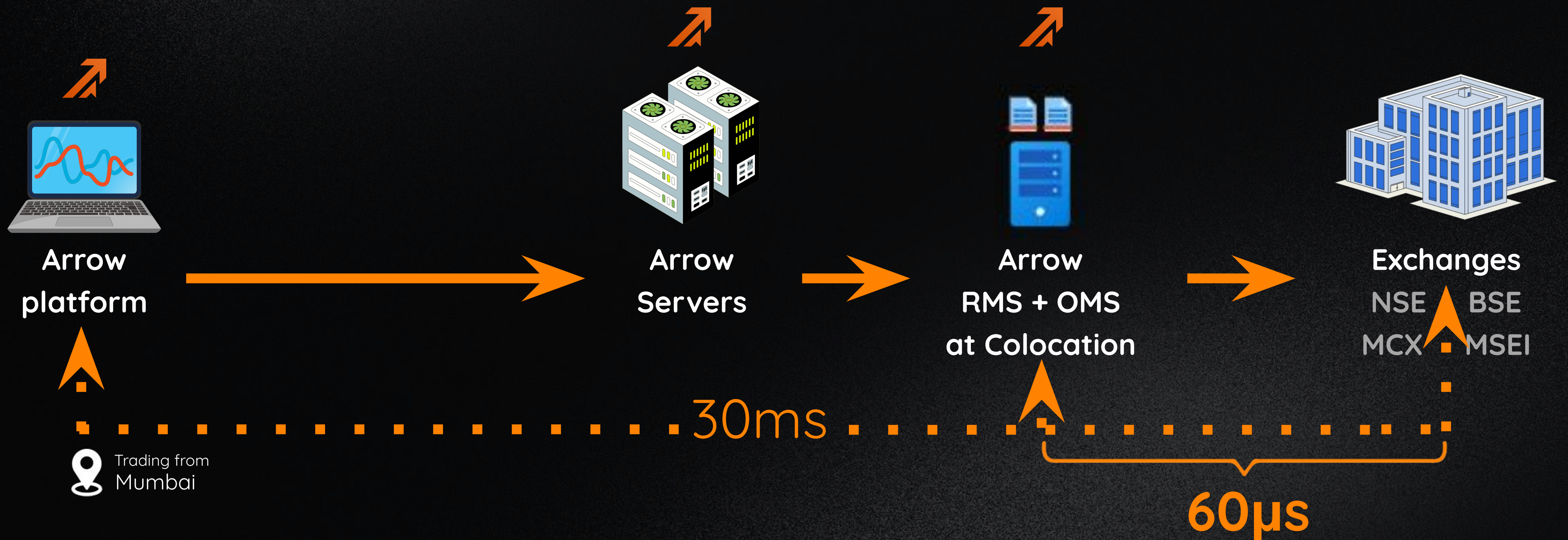
**Unlock Multiple Strategies**

**Risk Management**

# How do we do it?

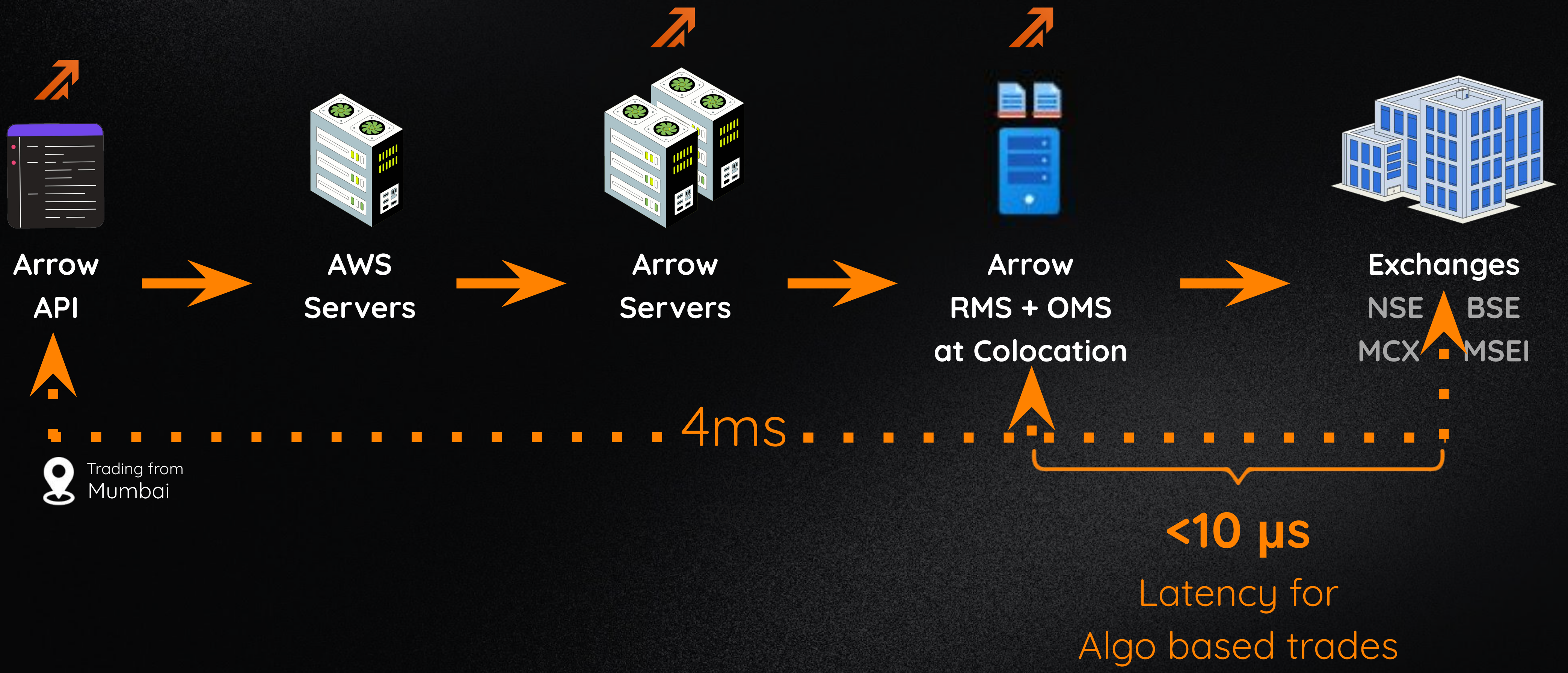
# Platform Trading

Execute trades with institutional-grade speed and precision through Arrow's advanced trading platform.



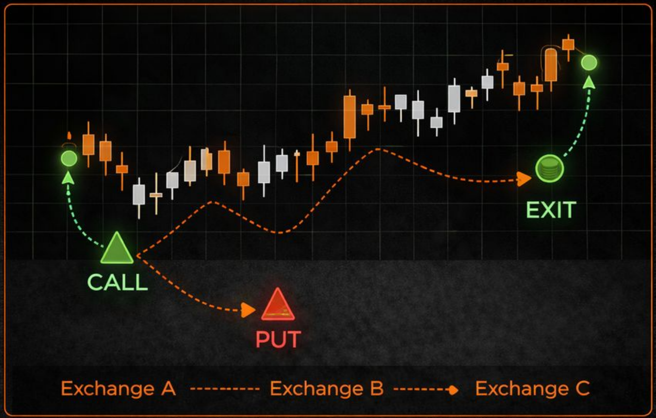
# API Trading

Integrate your strategies directly with Arrow's low-latency API for automated, high-performance trading.



# Unlock Strategies, like-

## Option Scalping

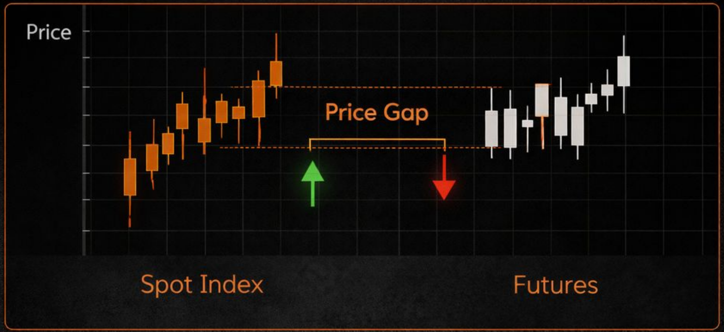



## Futures Rollovers

Seamlessly manage contract expirations.

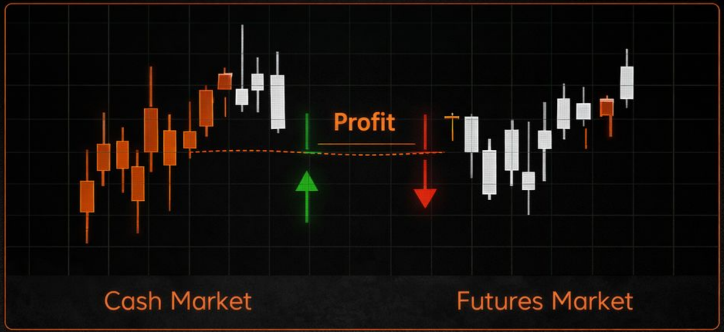

## Index Arbitrage

Exploit inefficiencies between spot index and futures.

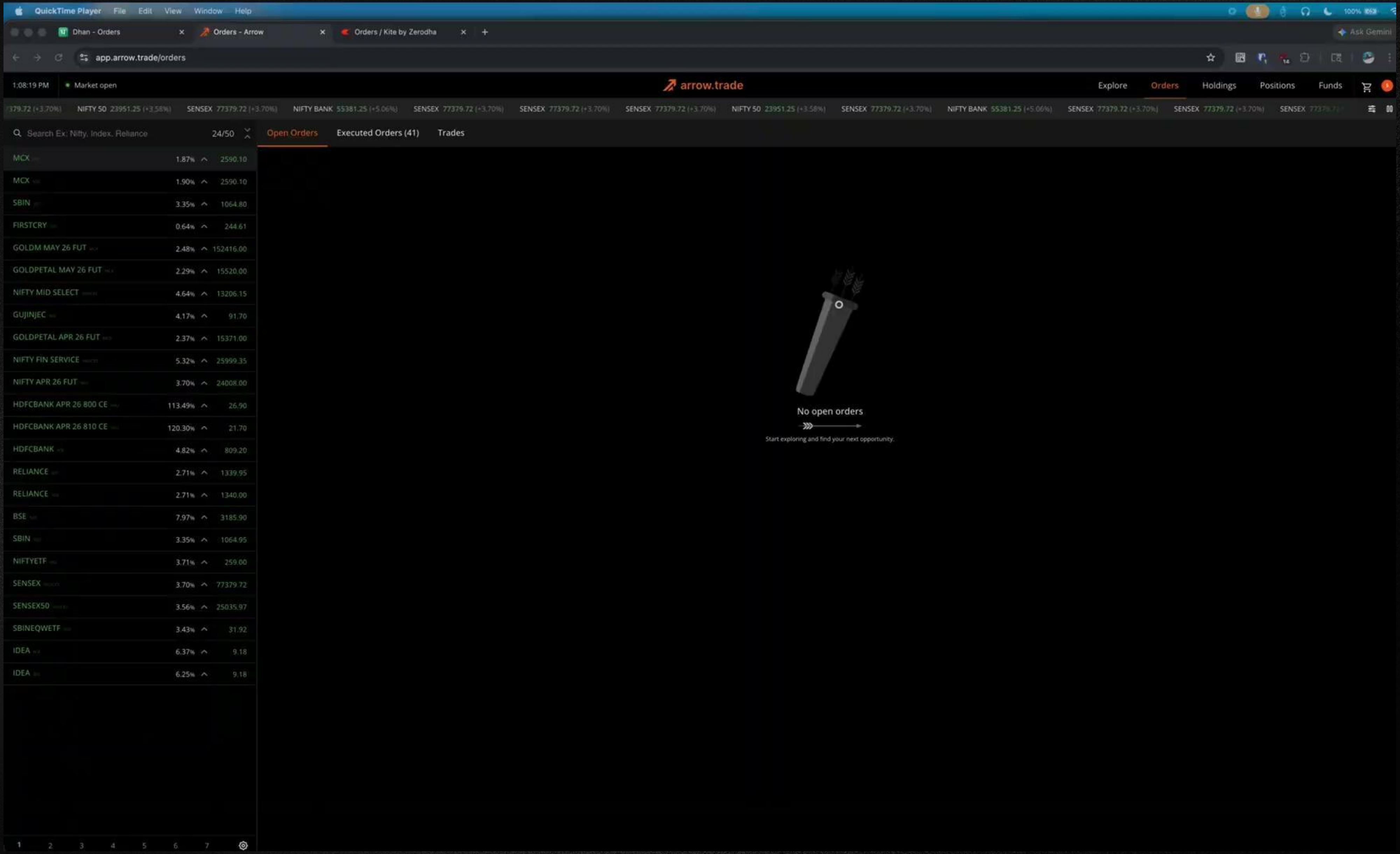



## Cash-Future Arbitrage

Profit from price gaps between cash and futures markets.



# Lets see a live trade



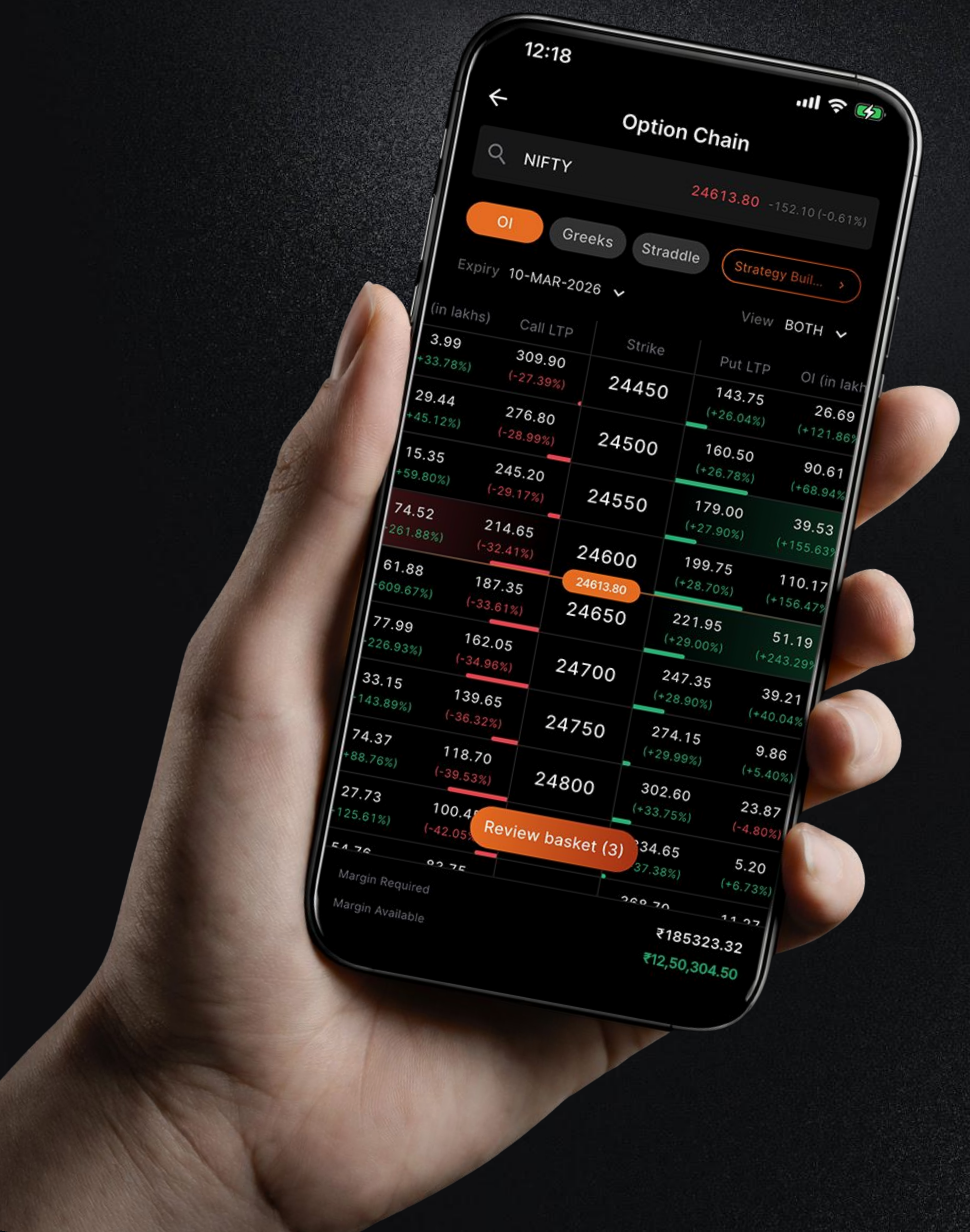
WE ARE NOT JUST ANOTHER BROKING PLATFORM.

**THIS IS INSTITUTIONAL INFRASTRUCTURE, DEMOCRATIZED.**

17 YEARS OF HFT EXPERIENCE,

**NOW IN YOUR HANDS.**

arrow.trade




# arrow.trade

Thank you for your time and attention.

Would Love to know about you!